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Robustness of CLIP-fusion classifiers under naturalistic and adversarial perturbations

Enric Guasch Mesia Henrik Paul Gruber Jane Klavir
Group 49

Abstract

Hateful meme detectors are deployed against users who actively obfuscate, yet are usually benchmarked only on clean inputs. We audit a CLIP-fusion classifier on meme datasets under naturalistic label-preserving edits and white-box adversarial attacks. The audit exposes two failures: text edits dominate degradation as the image branch is effectively dead (image-only AUROC 0.60 vs. a dedicated image baseline 0.65); and it is undefended against white-box PGD (AUROC < 0.01 already at $\epsilon=2/255$). Clean $\leftrightarrow$ perturbed KL-consistency plus per-example text-modality dropout revives the image branch (image-only AUROC 0.60 $\rightarrow$ 0.66, exceeding the dedicated baseline) and lifts the count of fully naturalistic-robust held-out examples by 76% at zero clean-accuracy cost — a class-asymmetric false-positive veto (97% of corrections on benign memes). Madry-style adversarial training in turn closes the white-box gap (PGD AUROC 0 $\rightarrow$ 0.62, stable under 100-step attacks) at a ≈ 0.12 clean-AUROC cost, and a combined model attains both robustness types at that cost.

Keywords: hateful memes, multimodal robustness, CLIP, consistency regularization, modality dropout.

1. Introduction

Automated hateful meme moderation is adversarial by construction: users could rewrite captions/ re-encode images to evade it, yet such models are reported almost exclusively on clean benchmark accuracy [1, 2]. We audit a CLIP-fusion classifier under two complementary threat models — naturalistic, label-preserving obfuscations and white-box image attacks — and ask whether a simple training-time intervention can close the naturalistic gap without sacrificing clean accuracy. Using a controlled perturbation suite (8 text $\times$ 11 image attacks $\times$ 3 severities, plus composites, deterministically seeded) over three seeds and three splits, we contribute: (i) an audit showing the model is a text shortcut with a dead image branch; text edits dominate naturalistic degradation

while white-box PGD breaks every recipe; (ii) a lightweight recipe (KL consistency plus text-modality dropout) that revives the image branch at zero clean-accuracy cost; and (iii) a failure analysis showing the defense chiefly prevents false positives on benign memes; and (iv) an adversarial-training study showing white-box robustness is attainable with a combined model robust to both threats.

2. Related Work

The Hateful Memes Challenge implements the multimodal nature of memes with benign confounders that flip the label when a single modality changes, exhibiting a model-human gap [1]. Recent systems exploit the shared CLIP embedding space [3]: Hate-CLIPper cross-correlates the image and text embeddings through a feature-interaction matrix [2], and MemeCLIP adds lightweight adapters and a cosine classifier for small, imbalanced meme sets [4]. These optimise clean benchmark accuracy and do not characterise behaviour under corruption/ attack. Cross-domain performance is often text-driven while image features are dataset-sensitive [5]. Robustness to label-preserving corruptions and white-box L_∞ attacks [6, 7, 8] is well studied for unimodal vision but rarely measured for CLIP-fusion meme detectors. We close this gap with one controlled suite that diagnoses the fragile modality and compares clean vs. robustness-aware training.

3. Method

Data. We use the Hateful Memes dataset [1]: 8,500 train / 500 dev memes, plus the labelled `test_seen` ($n=1000$) and `test_unseen` ($n=2000$) splits for held-out evaluation, keeping the official split structure for comparability. Complementarily, we use the MAMI [9] and MultiOFF [10] datasets, to show that our model translates well.

Classifier. A dual-encoder CLIP-fusion model on OpenCLIP ViT-B/32 maps the image and tokenized caption to $v, t \in \mathbb{R}^{512}$. Pixel normalization is a module inside the network, so the forward pass is differentiable with respect to raw $[0, 1]$ pixels (shared by white-box attacks and consistency training). A fusion head consumes

$$z = [t; v; |t - v|; t \odot v] \in \mathbb{R}^{2048} \quad (1)$$

where $|t - v|$ captures cross-modal disagreement and $t \odot v$

alignment, then applies $\text{LN} \rightarrow \text{Lin}(2048, 512) \rightarrow \text{GELU} \rightarrow \text{Lin}(512, 1)$.

Perturbation suite. Eight text attacks (e.g. leetspeak, char deletion, typos, etc.) and eleven image attacks (e.g. Gaussian noise/blur, JPEG compression, contrast shift, etc.) each have low/medium/high presets, with medium the realistic level reachable with an off-the-shelf editor and high a stress ceiling (see Figure 1). Cells use SHA-256-keyed per-example seeds; composite cells apply $K \in \{2, 4\}$ perturbations per sample to model a determined user.



Figure 1: Example Image: Original Image → Natural Perturbation (contrast down) → Adversarial Attack.

Robustness-aware training. For this part of training, both encoders are frozen and we are just training on the fusion head. Each batch pairs a clean view with an on-the-fly perturbed view (perturb text/image/both with prob. 0.5/0.3/0.2). **AugOnly** supervises both views, $\mathcal{L}_{\text{Aug}} = \text{BCE}(y, \ell_{\text{clean}}) + \alpha \text{BCE}(y, \ell_{\text{pert}})$. **KL** adds consistency to the detached clean prediction, $\mathcal{L}_{\text{KL}} = \mathcal{L}_{\text{Aug}} + \beta \text{KL}(p_{\text{clean}} \| p_{\text{pert}})$ with $p = \sigma(\ell)$, $\alpha=1$, $\beta=0.5$. **KLDrop** additionally zeros the text embedding for a fraction p of examples, forcing a real image-only prediction. The clean baseline minimizes BCE alone, with AdamW, cosine decay, and early stopping on dev macro-F1.

White-box attacks. With an unfrozen image encoder we probe the worst case with image-space FGSM [6] and PGD [7] (L_∞) over $\epsilon \in \{1, 2, 4, 8\}/255$. As the targeted remedy for this regime we implement Madry-style adversarial training and a TRADES-style [8] consistency variant; ϵ -values above 8/255 showed visible noise on the photo and were therefore ignored for the purpose of this project.

Metrics. We report AUROC (threshold-free, primary), macro-F1, and FPR/FNR (moderation has asymmetric costs). Per cell we report the robustness gap $\Delta \text{AUROC} = \text{AUROC}_{\text{clean}} - \text{AUROC}_{\text{att}}$ and attack success rate (ASR), with thresholds fixed on clean dev and reused under attack. Every recipe runs for 3 seeds (mean $\pm$ std).

4. Validation

We evaluate every recipe over 3 seeds on the test sets, against a majority baseline and dedicated unimodal classifiers; image and text-only heads trained from scratch.

Audit: a text shortcut with a dead image branch. Fusion beats every unimodal baseline (Table 1), but the gain is lop-

Model	AUROC	macro-F1	Acc.
Majority (all-0)	0.500	0.385	0.625
Image-only (ded.)	0.650	0.601	0.615
Text-only (ded.)	0.635	0.596	0.617
Multimodal fusion	0.738	0.667	0.697
<i>Fusion model, single-branch forward:</i>			
image-only fwd	0.602	0.563	0.591
text-only fwd	0.612	0.570	0.588

Table 1: Clean results on held-out `test_unseen`.

Recipe	Clean AUROC	W-text Δ	Comp Δ	Image-only	Robust cnt/2000
clean	0.738	0.110	0.073	0.602	222
AugOnly	0.708	0.063	0.043	0.620	282
KL	0.738	0.065	0.052	0.628	352
KLDrop-p015	0.743	0.069	0.054	0.658	391
KLDrop-p050	0.712	0.049	0.035	0.666	331

Table 2: Recipes on held-out `test_unseen` (3-seed mean).

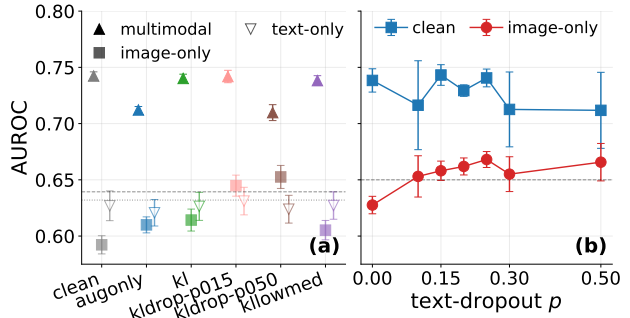


Figure 2: (a) Per-branch forward AUROC (mean over 3 splits $\times$ 3 seeds). (b) Dropout sweep (`test_unseen`).

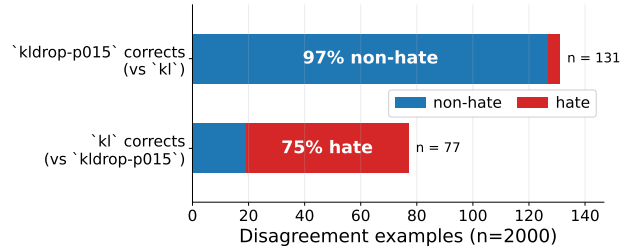


Figure 3: Per-example disagreements between KLDrop-p015 and KL on `test_unseen` ($n=2000$), split by gold label.

sided: its image-only forward (AUROC 0.60) sits roughly 0.05 below the dedicated image baseline (0.65) — about twice the deficit of the text branch — so the head leans on text, underusing the image. Naturalistic degradation is thus text-driven. This holds on all three splits (Fig. 2a); the model is more text-fragile than the dedicated text baseline.

Defense: reviving the image branch at zero clean cost. The dropout sweep (Fig. 2b) shows image-only AUROC rises monotonically with p , peaking at 0.668 ($p=0.25$), while multimodal AUROC stays within seed noise through $p \leq 0.25$. $p=0.15$ (KLDrop-p015) is the default, and $p=0.50$ is a robustness-heavy alternative. KLDrop redistributes capacity rather than adding signal (Fig. 2a): it

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Model	Clean AUROC	Naturalistic AUROC (Δ)	White-box PGD AUROC (Δ)
Clean	0.738	0.598 (0.14)	< 0.01 (0.73)
Naturalistic	0.743	0.643 (0.10)	< 0.01 (0.73)
Adversarial (clean-pres)	0.741	0.590 (0.15)	0.531 (0.21)
Adversarial (robust)	0.616	0.540 (0.08)	0.625 (≈ 0)
Combined	0.614	0.548 (0.07)	0.610 (≈ 0)

Table 3: Cross-robustness on `test_unseen` (3-seed mean AUROC): clean, and attacked AUROC (drop Δ vs. clean) under worst naturalistic cell and white-box PGD (L_∞ , $\epsilon=8/255$, 25 steps).

lifts its image-only forward to or above the dedicated baseline on every split, leaving text-only unchanged. Table 2 gives the held-out trade-off: KL preserves clean AUROC, KLDrop-p015 even nudges it up while improving every robustness column. The number of memes correctly withstanding all 57 naturalistic permutations rises from $222 \rightarrow 391$ (+76%). Composites are the realistic threat: two stacked text edits cost the clean model $\Delta\text{AUROC} = 0.073$ (medium) to 0.111 (high) on `composite_2text` — on par with its worst single cell, yet reachable with everyday edits — and KLDrop roughly halves the medium gap ($0.073 \rightarrow 0.035$, p050).

Why it works: a class-asymmetric false-positive veto. A per-example analysis on `test_unseen` reveals the mechanism (Fig. 3): where KLDrop-p015 and KL disagree, 97% of KLDrop’s corrections are on benign memes (95% CI [94, 99], $n=131$) vs. 75% on hateful memes for KL. The revived image branch vetoes the text branch when an edited caption looks superficially hateful but the image does not support it — the over-moderation failure that erodes trust. The choice is thus deployment-conditional: false-positive-sensitive contexts prefer KLDrop, recall-sensitive ones prefer KL; the direction holds on all three splits.

Adversarial Results. White-box PGD breaks every undefended model: clean and naturalistic (KLDrop) AUROC collapse from ≈ 0.74 to < 0.01 ($\text{ASR} > 0.98$) already at $\epsilon=2/255$ (Table 3). Adversarial training — unfreezing the vision encoder — restores robustness but exposes a clean-robustness trade. A clean-preserving recipe (TRADES, early-stopped) keeps clean ≈ 0.74 but its robustness does not generalize: under PGD on held-out `test_unseen` it falls to chance (0.53 AUROC, ASR 0.26 at 25 steps; 0.26 AUROC at 100). A robustness-first recipe (Madry) instead holds ≈ 0.62 AUROC under PGD with $\text{ASR} \approx 0$, flat across 10/25/100 steps at a ≈ 0.12 clean-AUROC cost. As white-box access is unrealistic for a server-side detector, this cost is worth paying only when model weights are exposed.

Combined model. Training jointly on clean, naturalistic and PGD views under one composite loss $\mathcal{L}_{\text{comb}} = \mathcal{L}_{\text{KL}} + \delta \text{BCE}(y, \ell_{\text{adv}})$ ($\delta=1$, KLDrop text-dropout in the forward, ℓ_{adv} the logit of the PGD view) yields one model robust on both axes: the smallest degradation under both the worst naturalistic cell ($\Delta 0.07$) and PGD ($\Delta \approx 0$, ASR 0), at the same ≈ 0.12 clean cost as adversarial training. Warm-

starting the combined model from the clean-preserving adversarial checkpoint (encoder frozen) instead recovers only the naturalistic operating point (clean 0.75, PGD random), confirming the trade-off: high clean accuracy and generalizing white-box robustness cannot coexist. The combined model is hence the choice when both threats matter and some clean accuracy can be spent; otherwise KLDrop is the practical default.

Applicability to other Datasets. As further validation, we applied our training to datasets MAMI and MultiOFF, yielding similar and consistent preliminary results, proving that our approach translates well to other datasets.

Limitations. In our experiments, we reasonably assume perturbations are label preserving. To guarantee this assumption holds, every modified meme would need inspection, which is beyond the scope of the project. The model was fine tuned on hate speech specifically; similar domains (e.g. misinformation or manipulation) are out of scope.

5. Conclusion

A CLIP-fusion hateful meme detector is a text shortcut with a dead image branch — fragile to the text edits a real user applies, and undefended against white-box PGD. Our recipe (KL consistency plus text-modality dropout) revives the image branch above a dedicated baseline and raises held-out naturalistic robustness by 76% at zero clean-accuracy cost, acting as a class-asymmetric false-positive veto for benign content; the white-box training substantially robustifies against adversarial attacks. Hateful meme detectors should thus be reported with a modality-resolved robustness audit, not clean accuracy alone, since the dominant failure — over-flagging benign speech after a trivial edit — is invisible on clean benchmarks. These are research prototypes, not moderators; we release the suite for defensive evaluation.

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